Tarea M7-35

Alumno: Yamil Salomón

Práctica: Introducción al lenguaje SQL

1. Configuración Inicial

Para asegurar la ejecución correcta del script, se definió la base de datos de trabajo.

- Comando utilizado: USE AdventureWorks2022

2. Desarrollo de Consultas (DML)

Punto A: Uso de SELECT (Consulta General)

- Objetivo: Recuperar la totalidad de registros de la tabla de empleados para una auditoría inicial.

```sql
/* ==================================================
TAREA M7 - YAMIL SALOMON
Módulo 35: Introducción al lenguaje SQL
==================================================
*/

-- 1. CONFIGURACIÓN: Establecer la base de datos de trabajo
USE AdventureWorks2022;
GO

-- 2. REQUERIMIENTO: Comando SELECT [cite: 372]
-- Objetivo: Consultar la información básica de todos los empleados.
-- Interpretación: Este comando permite al analista tener una visión
-- general de la plantilla para auditorías iniciales.
SELECT * FROM HumanResources.Employee;
```

Results | Messages

	BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours	CurrentFlag	rowguid	ModifiedDate
1	1	295847284	adventure-works\ken0	NULL	NULL	Chief Executive Officer	1969-01-29	S	M	2009-01-14	1	99	69	1	F01251E5-96A3-448D-981E-0F99D789110D	2014-06-30 00:0
2	2	245797967	adventure-works\terri0	0x58	1	Vice President of Engineering	1971-08-01	S	F	2008-01-31	1	1	20	1	45E8F437-670D-4409-93CB-F9424A40D6EE	2014-06-30 00:0
3	3	509647174	adventure-works\roberto0	0x5AC0	2	Engineering Manager	1974-11-12	M	M	2007-11-11	1	2	21	1	9BBBFB2C-EFBB-4217-9AB7-F97689328841	2014-06-30 00:0
4	4	112457891	adventure-works\rob0	0x5AD6	3	Senior Tool Designer	1974-12-23	S	M	2007-12-05	0	48	80	1	59747955-87B8-443F-8ED4-F8AD3AFDF3A9	2014-06-30 00:0
5	5	695256908	adventure-works\gail0	0x5ADA	3	Design Engineer	1952-09-27	M	F	2008-01-06	1	5	22	1	EC84AE09-F9B8-4A15-B4A9-6CCBAB919B08	2014-06-30 00:0
6	6	998320692	adventure-works\jossef0	0x5ADE	3	Design Engineer	1959-03-11	M	M	2008-01-24	1	6	23	1	E39056F1-9CD5-478D-8945-14ACA7FBDCDD	2014-06-30 00:0
7	7	134969118	adventure-works\dylan0	0x5AE1	3	Research and Development Manager	1987-02-24	M	M	2009-02-08	1	61	50	1	4F46DECA-EF01-41FD-9829-0ADAB368E431	2014-06-30 00:0
8	8	811994146	adventure-works\diane1	0x5AE158	4	Research and Development Engineer	1986-06-05	S	F	2008-12-29	1	62	51	1	31112635-663B-4018-B4A2-A685C0BF48A4	2014-06-30 00:0
9	9	658797903	adventure-works\gigi0	0x5AE168	4	Research and Development Engineer	1979-01-21	M	F	2009-01-16	1	63	51	1	50B6CDC6-7570-47EF-9570-48A64B5F2ECF	2014-06-30 00:0
10	10	879342154	adventure-works\michael6	0x5AE178	4	Research and Development Manager	1984-11-30	M	M	2009-05-03	1	16	64	1	EAA43680-5571-40CB-AB1A-3BF68F04459E	2014-06-30 00:0
11	11	974026903	adventure-works\ovidiu0	0x5AE3	3	Senior Tool Designer	1978-01-17	S	M	2010-12-05	0	7	23	1	F68C7C19-FAC1-438C-9BB7-AC33FCC341C3	2014-06-30 00:0
12	12	480168528	adventure-works\thierry0	0x5AE358	4	Tool Designer	1959-07-29	M	M	2007-12-11	0	9	24	1	1D955171-E773-4FAD-8382-40FD898D5D4D	2014-06-30 00:0
13	13	486228782	adventure-works\janice0	0x5AE368	4	Tool Designer	1989-05-28	M	F	2010-12-23	0	8	24	1	954B91B6-5AA7-48C2-8685-6E11C6E5C49A	2014-06-30 00:0
14	14	42487730	adventure-works\michael8	0x5AE5	3	Senior Design Engineer	1979-06-16	S	M	2010-12-30	1	3	21	1	46286CA4-46DD-4DDB-9128-85B67E98D1A9	2014-06-30 00:0
15	15	56920285	adventure-works\sharon0	0x5AE7	3	Design Engineer	1961-05-02	M	F	2011-01-18	1	4	22	1	54F2FDC0-87C4-4065-A7A8-9AC8EA624235	2014-06-30 00:0
16	16	24756624	adventure-works\david0	0x68	1	Marketing Manager	1975-03-19	S	M	2007-12-20	1	40	40	1	E87029AA-2CBA-4C03-B948-D83AF0313E28	2014-06-30 00:0
17	17	253022876	adventure-works\kevin0	0x6AC0	2	Marketing Assistant	1987-05-03	S	M	2007-01-26	0	42	41	1	1B480240-95C0-410F-A717-EB29943C8886	2014-06-30 00:0
18	18	222969461	adventure-works\john5	0x6B40	2	Marketing Specialist	1978-03-06	S	M	2011-02-07	0	48	44	1	64730415-1F58-4E5B-8FA8-5E4DAEBA53B4	2014-06-30 00:0
19	19	52541318	adventure-works\mary2	0x6BC0	2	Marketing Assistant	1978-01-29	S	F	2011-02-14	0	43	41	1	1F6DA901-C7F7-48A8-8EEF-D81868D72B52	2014-06-30 00:0
20	20	323403273	adventure-works\wanida0	0x6C20	2	Marketing Assistant	1975-03-17	M	F	2011-01-07	0	41	40	1	43CCA446-DA1C-454C-8530-873AD2923E1B	2014-06-30 00:0
21	21	243322160	adventure-works\terry0	0x6C60	2	Marketing Specialist	1986-02-04	M	M	2009-03-02	0	44	42	1	AD0D7626-4506-45C4-898F-9D7A0E13AABD	2014-06-30 00:0
22	22	95958330	adventure-works\sariya0	0x6CA0	2	Marketing Specialist	1987-05-21	S	M	2008-12-12	0	45	42	1	AF21F1B7-0691-48AD-B325-B8F2D7B2268A	2014-06-30 00:0
23	23	767955365	adventure-works\mary0	0x6CE0	2	Marketing Specialist	1962-09-13	M	F	2009-01-12	0	46	43	1	90899830-127E-4876-85CA-7F6E03025281	2014-06-30 00:0
24	24	72636981	adventure-works\jill0	0x6D10	2	Marketing Specialist	1979-06-18	M	F	2009-01-18	0	47	43	1	53575C58-FBA1-4C1A-B8FF-95E050B44393	2014-06-30 00:0
25	25	519899904	adventure-works\james1	0x78	1	Vice President of Production	1983-01-07	S	M	2009-02-03	1	64	52	1	D7314F24-2AF1-429C-9BBB-4038F45F3E6C	2014-06-30 00:0
26	26	277173473	adventure-works\peter0	0x7AC0	2	Production Control Manager	1982-11-03	M	M	2008-12-01	1	43	41	1	69D5D162-E817-45E7-9DEC-5D9B8310E7B1	2014-12-26 09:1
27	27	446466105	adventure-works\jo0	0x7AD6	3	Production Supervisor - WC60	1956-10-08	S	F	2008-02-27	0	80	60	1	83FFAAC6-B895-481F-B897-14F965D4DA47	2014-06-30 00:0
28	28	14417807	adventure-works\guy1	0x7AD6B0	4	Production Technician - WC60	1988-03-13	M	M	2006-06-30	0	21	30	1	AAE1D04A-C237-4974-B4D5-935247737718	2014-06-30 00:0
29	29	948320468	adventure-works\mark1	0x7AD6D0	4	Production Technician - WC60	1979-09-25	S	M	2009-01-23	0	19	29	1	CDB2C6DF-93E5-4558-91C8-424D5E7C658B	2014-06-30 00:0
30	30	410742000	adventure-works\britta0	0x7AD6F0	4	Production Technician - WC60	1989-09-28	M	F	2009-01-29	0	14	27	1	FC12677F-42F8-4790-AD69-685088ED7E5F	2014-06-30 00:0
31	31	750246141	adventure-works\margie0	0x7AD708	4	Production Technician - WC60	1986-05-20	M	F	2009-01-04	0	18	29	1	352D6E2F-655B-49A2-9FEC-EFD7D8CD0D30	2014-06-30 00:0
32	32	330211482	adventure-works\rebecca0	0x7AD718	4	Production Technician - WC60	1977-07-10	M	F	2008-12-29	0	23	31	1	BDADEB5C-A596-4DA1-88BA-842C7A0CC10F	2014-06-30 00:0

Query executed successfully. | HUAWEID15-YAMO (17.0 RTM) | HUAWEID15-YAMO\Yamil (63) | AdventureWorks2022 | 00:00:00 | Row: 1, Col: 1 | 290 rows

- Interpretación de Negocio: Este comando es el punto de partida para cualquier análisis exploratorio de datos (EDA), permitiendo entender qué columnas y tipos de información tenemos disponibles en la nómina.

Punto B: Uso de WHERE (Filtrado Específico)

- Objetivo: Identificar únicamente al personal con el cargo de 'Design Engineer'.

```sql
-- 3. REQUERIMIENTO: Comando WHERE
-- Objetivo: Filtrar empleados que tienen el título de 'Design Engineer'.
-- Interpretación: Permite al departamento de ingeniería identificar
-- perfiles específicos para asignación de proyectos técnicos.
SELECT * FROM HumanResources.Employee
WHERE JobTitle = 'Design Engineer';
```

	BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours
1	5	695256908	adventure-works\gail0	0x5ADA	3	Design Engineer	1952-09-27	M	F	2008-01-06	1	5	22
2	6	998320692	adventure-works\jossef0	0x5ADE	3	Design Engineer	1959-03-11	M	M	2008-01-24	1	6	23
3	15	56920285	adventure-works\sharon0	0x5AE7	3	Design Engineer	1961-05-02	M	F	2011-01-18	1	4	22

- Interpretación de Negocio: La segmentación precisa permite optimizar la comunicación interna, dirigiendo mensajes o capacitaciones solo al personal técnico relevante, evitando el "ruido" informativo en otros departamentos.

Punto C: Uso de BETWEEN (Rangos de Datos)

- Objetivo: Filtrar empleados con un rango de horas de enfermedad entre 40 y 80.

- Interpretación de Negocio: Como analista, este filtro ayuda a detectar anomalías en el bienestar laboral. Un rango elevado podría indicar la necesidad de revisar las cargas de trabajo o las condiciones de salud en áreas específicas.

```
-- 4. REQUERIMIENTO: Comando BETWEEN
-- Objetivo: Identificar empleados con horas de enfermedad (SickLeaveHours) entre 40 y 80.
-- Interpretación: Herramienta de control para RH para monitorear el
-- bienestar del personal y detectar posibles ausentismos recurrentes.
SELECT * FROM HumanResources.Employee
WHERE SickLeaveHours BETWEEN 40 AND 80;
```

	BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	HireDate	SalariedFlag	VacationHours	SickLeaveHours
1	1	295847284	adventure-works\ken0	NULL	NULL	Chief Executive Officer	1969-01-29	S	M	2009-01-14	1	99	69
2	4	112457891	adventure-works\rob0	0x5AD6	3	Senior Tool Designer	1974-12-23	S	M	2007-12-05	0	48	80
3	7	134969118	adventure-works\dylan0	0x5AE1	3	Research and Development Manager	1987-02-24	M	M	2009-02-08	1	61	50
4	8	811994146	adventure-works\diane1	0x5AE158	4	Research and Development Engineer	1986-06-05	S	F	2008-12-29	1	62	51
5	9	658797903	adventure-works\gigi0	0x5AE168	4	Research and Development Engineer	1979-01-21	M	F	2009-01-16	1	63	51
6	10	879342154	adventure-works\mich...	0x5AE178	4	Research and Development Manager	1984-11-30	M	M	2009-05-03	1	16	64
7	16	24756624	adventure-works\david0	0x68	1	Marketing Manager	1975-03-19	S	M	2007-12-20	1	40	40
8	17	253022876	adventure-works\kevin0	0x6AC0	2	Marketing Assistant	1987-05-03	S	M	2007-01-26	0	42	41
9	18	222969461	adventure-works\john5	0x6B40	2	Marketing Specialist	1978-03-06	S	M	2011-02-07	0	48	44
10	19	52541318	adventure-works\mary2	0x6BC0	2	Marketing Assistant	1978-01-29	S	F	2011-02-14	0	43	41
11	20	323403273	adventure-works\wani...	0x6C20	2	Marketing Assistant	1975-03-17	M	F	2011-01-07	0	41	40
12	21	243322160	adventure-works\terry0	0x6C60	2	Marketing Specialist	1986-02-04	M	M	2009-03-02	0	44	42
13	22	95958330	adventure-works\sariya0	0x6CA0	2	Marketing Specialist	1987-05-21	S	M	2008-12-12	0	45	42
14	23	767955365	adventure-works\mary0	0x6CE0	2	Marketing Specialist	1962-09-13	M	F	2009-01-12	0	46	43
15	24	72636981	adventure-works\jill0	0x6D10	2	Marketing Specialist	1979-06-18	M	F	2009-01-18	0	47	43
16	25	519899904	adventure-works\jame...	0x78	1	Vice President of Production	1983-01-07	S	M	2009-02-03	1	64	52
17	26	277173473	adventure-works\peter0	0x7AC0	2	Production Control Manager	1982-11-03	M	M	2008-12-01	1	43	41
18	27	446466105	adventure-works\jo0	0x7AD6	3	Production Supervisor - WC60	1956-10-08	S	F	2008-02-27	0	80	60
19	40	309738752	adventure-works\jolynn0	0x7ADA	3	Production Supervisor - WC60	1956-01-16	S	F	2007-12-26	0	82	61

Punto D: Uso de ORDER BY (Ordenamiento)

- Objetivo: Organizar los registros de forma ascendente por fecha de nacimiento

```
-- 5. REQUERIMIENTO: Comando ORDER BY
-- Objetivo: Listar a todos los empleados ordenados por su fecha de nacimiento (antigüedad de edad).
-- Interpretación: Útil para planeación de jubilaciones o programas
-- de beneficios por rangos generacionales.
SELECT * FROM HumanResources.Employee
ORDER BY BirthDate ASC;
```

	BusinessEntityID	NationalIDNumber	LoginID	OrganizationNode	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender
1	274	502097814	adventure-works\stephen0	0x9560	2	North American Sales Manager	1951-10-17	M	M
2	84	947029962	adventure-works\frank3	0x7AE794	4	Production Technician - WC40	1952-03-02	M	M
3	256	367453993	adventure-works\frank2	0x85EB94	4	Buyer	1952-05-12	M	M
4	5	695256908	adventure-works\gail0	0x5ADA	3	Design Engineer	1952-09-27	M	F
5	140	339233463	adventure-works\prasan...	0x7AEECA	4	Production Technician - WC20	1953-04-30	M	M
6	231	153479919	adventure-works\jo1	0x7C2B78	4	Janitor	1954-04-24	M	F
7	63	414476027	adventure-works\maciej0	0x7AE358	4	Production Technician - WC60	1955-01-30	S	M
8	40	309738752	adventure-works\jolynn0	0x7ADA	3	Production Supervisor - WC60	1956-01-16	S	F
9	49	912265825	adventure-works\barry0	0x7ADED0	4	Production Technician - WC10	1956-03-26	S	M
10	53	9659517	adventure-works\diane0	0x7ADF28	4	Production Technician - WC10	1956-03-29	M	F
11	55	233069302	adventure-works\taylor0	0x7AE1	3	Production Supervisor - WC50	1956-04-01	M	M
12	52	565090917	adventure-works\doris0	0x7ADF18	4	Production Technician - WC10	1956-04-04	M	F
13	48	690627818	adventure-works\ruth0	0x7ADEB0	4	Production Technician - WC10	1956-06-04	M	F
14	51	132674823	adventure-works\jeffrey0	0x7ADF08	4	Production Technician - WC10	1956-07-11	S	M
15	62	494170342	adventure-works\john0	0x7AE3	3	Production Supervisor - WC60	1956-08-07	M	M
16	50	844973625	adventure-works\sidney0	0x7ADEF0	4	Production Technician - WC10	1956-08-30	M	M

- Interpretación de Negocio: El ordenamiento es clave para reportes ejecutivos de antigüedad y planeación de recursos humanos a largo plazo (como planes de jubilación).